Tianhao Wu

Email: wutianha@usc.edu **Webpage**: https://tianhao-stan-wu.github.io/

Research interests

My research interests lie at the intersection of formal methods, cyber-physical systems and safe autonomy. I am interested in developing expressive spatio-temporal logics and quantitative semantics, designing verifiable safety frameworks for autonomous systems under uncertainty, and validating them on safety-critical applications such as autonomous driving.

Education

University of Southern California
M.S. in Computer Science — Aug 2025 – Present

B.S. in Computer Science and Computer Engineering — Jan 2021 – Dec 2024
B.S. in Applied and Computational Mathematics — Jan 2021 – Dec 2024
GPA: 3.95/4.0

Publications

An Algebraic Framework for Quantitative Semantics of Spatio-Temporal Logic with Graph Operators
Sheryl Paul*, Vidisha Kudalkar*, Anand Balakrishnan*, **Tianhao Wu**, Lars Lindemann, Jyotirmoy V. Deshmukh
QEST + FORMATS 2026

Feasibility Restoration under Conflicting STL Specifications with Pareto-Optimal Refinement
Tianhao Wu, Yiwei Lyu
arXiv 2026

Enhancing the Performance of DeepReach on High-Dimensional Systems through Optimizing Activation Functions
Qian Wang*, **Tianhao Wu***
arXiv 2023

Research experience

Research assistant, University of Southern California
Advisor: Jyotirmoy V. Deshmukh — Apr 2026 – Present
Algebraic Semantics of STL-GO, MARL STL-GO

Research assistant, Texas A&M University
Advisor: Yiwei Lyu — Oct 2025 – Present
Spatial-temporal STL relaxation in uncontrollable agents with multi-objective optimization

Research assistant, Carnegie Mellon University

* Indicates equal contribution.

Advisor: John M. Dolan — Jan 2025 – April 2025

Designed a safety framework for autonomous driving integrating vision-language models, Linear Temporal Logic, and Control Barrier Functions.

Research intern, SmartMore Cooperation

Advisor: Yukang Chen — May 2024 – Aug 2024

Investigated how the number and placement of cross-attention layers affect the performance of the LLaVA vision-language model.

Research assistant, University of Southern California

Advisor: Somil Bansal — Aug 2022 – May 2023

Evaluated DeepReach for Hamilton–Jacobi reachability on high-dimensional systems, proposing a modified activation architecture that reduced violation rates by 5.1% on a 9D multi-vehicle collision avoidance task.

Research assistant, University of California, Irvine

Advisor: Marco Levorato — Jun 2022 - Aug 2022

Developed an autonomous drone with obstacle detection and avoidance by training a perception model and integrating the embedded system.

Honors & Awards

Ming Hsieh Institute Undergraduate Scholar — 2023 – 2024

One of five scholars selected for academic excellence and research potential

USC Academic Achievement Award — Fall 2023

Viterbi CURVE Research Fellowship — 2022 – 2023

Lenore B. Kreiger Scholarship for Mathematics — 2022 – 2023

Teaching Experience

Teaching Assistant, University of Southern California

Fundamentals of Computation (CS102), Head TA — Spring 2022 – Spring 2024

Introduction to Embedded Systems (EE109) — Fall 2023

Introduction to Artificial Intelligence (CS360) — Spring 2023

CS@SC Summer Coding Camp — Summer 2022

Skills

Programming: Python, C++, MATLAB

Tools: Linux, LaTeX, Git, SSH, CARLA Simulator